

Research Mentorship Appreciation

Prof. Maria Giovanna Dainotti (NAOJ)



I first contacted Prof. Maria Giovanna Dainotti by email in March 2025 regarding the possibility of doing an internship under her guidance. She responded promptly and kindly connected me with her research group. From there, we started working on a project focused on the reconstruction of gamma-ray burst (GRB) light curves.

From the beginning, Prof. Dainotti was very supportive and actively involved in the research. We had regular weekly meetings where I presented my results, discussed the problems I was facing, and received her comments and suggestions. These meetings helped me understand how research is actually developed step by step and taught me to think more carefully about the methods I was using and the results I was obtaining.

During the first project, I worked on machine-learning and statistical approaches for reconstructing missing sections of GRB light curves. This gave me the opportunity to work different models including Deep Gaussian Processes, with noisy, irregular, and incomplete astronomical data and to explore different approaches to the same research problem. The project ultimately resulted in a paper on which I am a co-author, published in the Journal of High Energy Astrophysics (JHEAp, DOI: [10.1016/j.jheap.2025.100519](https://doi.org/10.1016/j.jheap.2025.100519)).

After completing and submitting the first paper, we continued our collaboration and started working on a new project related to GRB X-ray light-curve reconstruction using machine learning models. This work was also carried out under Prof. Dainotti's supervision. During this project, I worked on different machine-learning and statistical models, including ReFANN, Siamese Neural Networks. I also got the opportunity to supervise and guide other students working on the project. This gave me experience in explaining the methods I was using, discussing their results, and helping them when they faced problems. I also developed the ReFANN and Siamese Neural Network models from scratch and contributed to other models as well, including uncertainty estimation, model comparison, validation, and understanding the limitations of different approaches. This continued work has resulted in another research paper/preprint, currently submitted to a journal and also submitted to ([arXiv](https://arxiv.org)). We also submitted a contribution to the Proceedings of Science three weeks ago for a conference, and we are continuing to develop this research further.

One of the most valuable parts of working with Prof. Dainotti has been the regular research discussions. In our weekly meetings, I was encouraged to present not only successful results but also problems, unexpected results, and approaches that did not work. Her questions and comments helped me learn how to critically evaluate my own work

rather than simply focusing on obtaining a good result. I learned how to ask meaningful research questions, incorporate feedback, improve my analysis, and communicate scientific ideas more clearly.

Prof. Dainotti has also supported me beyond the research projects themselves. She helped and guided me during my summer internship and introduced me to other researchers in the field, including Prof. Dieter Hartmann, who works in high-energy astrophysics and gamma-ray bursts. These interactions helped me understand the broader research community and build connections with researchers working in related areas.

She is also currently supporting and guiding me as I prepare my PhD applications. Her advice on research, academic opportunities, and my future research direction has been very valuable to me as I move towards the next stage of my academic career.

Looking back, working with Prof. Dainotti has been much more than working on a research project. It has taught me how to approach an open research problem, work independently while knowing when to ask for guidance, deal with unexpected or negative results, and communicate my ideas clearly. It has also significantly shaped my research direction, particularly my interest in using machine learning and statistical methods to solve scientific problems.

I am sincerely grateful to Prof. Dainotti for her mentorship, guidance, patience, and encouragement throughout our collaboration. Her continuous support has significantly shaped my research direction and has given me greater confidence as an early-career researcher. I am very thankful for the opportunity to work with her and her research group, and for everything I have learned from this experience.

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2026